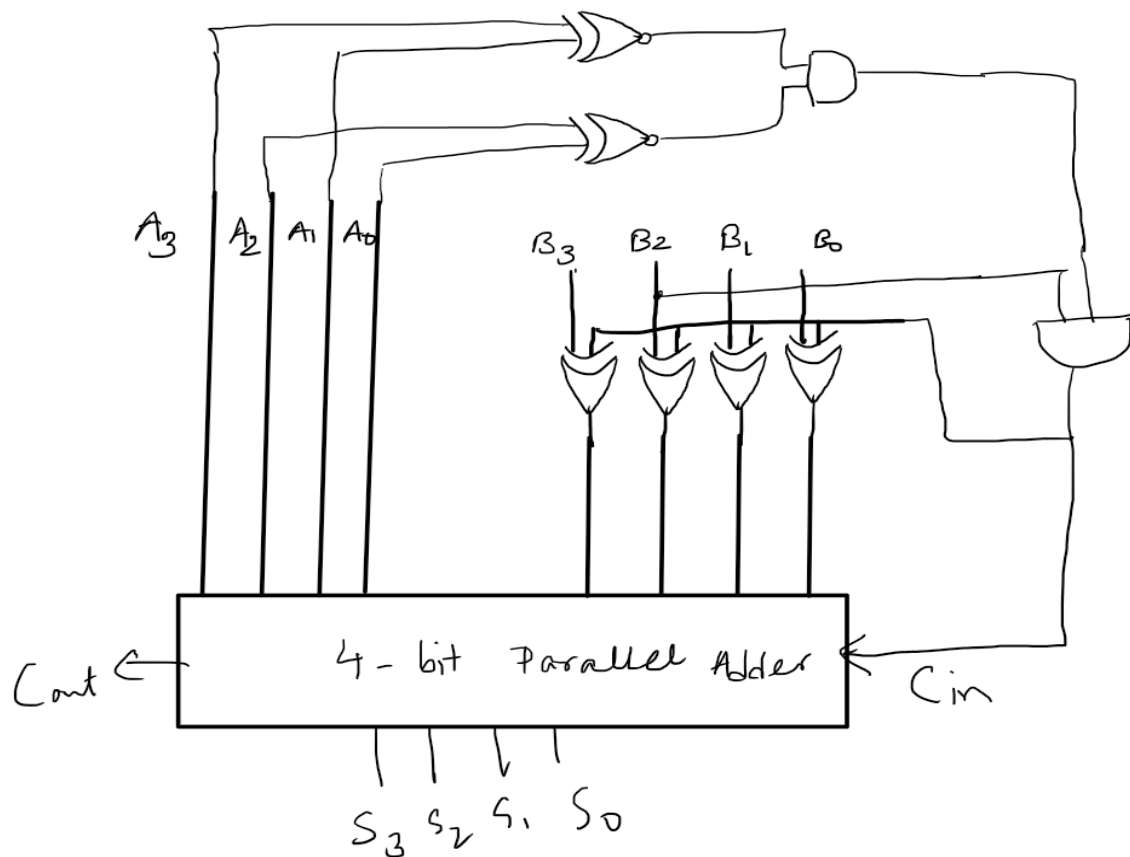


Name:	ID:	Section:
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1.	Design a circuit diagram for the following system that takes two 4-bit binary numbers [A and B] as inputs and outputs in the following fashion: If A is divisible by 5 and B is greater than 3, then the output should be A-B; otherwise perform A+B	10
2.	A memory has a total capacity of 3 MB, and each word is 24 bits. a. How many words does the memory hold? b. How many flip-flops are needed in the MBR? c. How many address lines are needed? d. How many flip-flops are required in the MAR? e. What size decoder would be needed to construct this RAM?	5

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2.

a. No. of words = Capacity/ bits per word

$$= 3 \times 1024 \times 1024 \times 8 / 24 = 1048576 \text{ words}$$

b. 24

c. [no.of](#) words = $2^{(\text{address lines})}$

$$\text{address lines} = \log_2 (1048576) = 20$$

d. 20

e. 20x1048576-sized decoder